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Research article

Modeling seasonal water yield for landscape management: Applications in Peru and Myanmar

Perrine Hamel [a,*,1], Jefferson Valencia [b], Rafael Schmitt [a], Manish Shrestha [c], Thanapon Piman [c], Richard P. Sharp [a], Wendy Francesconi [d], Andrew J. Guswa [e]

[a] The Natural Capital Project, Woods Institute for the Environment, Stanford University, Stanford, USA
[b] International Center for Tropical Agriculture (CIAT), Km 17 Recta Cali-Palmira. Z.C. 763537 - A.A., 6713, Cali, Colombia
[c] Stockholm Environment Institute (SEI), Asia Centre, Bangkok, Thailand
[d] International Center for Tropical Agriculture (CIAT), Av. La Molina, 1895, La Molina, Lima, Peru
[e] Picker Engineering Program, Smith College, Northampton, MA, 01063, USA

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ABSTRACT

A common objective of watershed management programs is to secure water supply, especially during the dry season. To develop such programs in contexts of low data and resource availability, program managers need tools to understand the effect of landscape management on the seasonal water balance. However, the performance of simple, parsimonious models is poorly understood. Here, we examine the behavior of a geospatial tool, developed to map monthly water budgets and baseflow contributions and forming part of the InVEST (integrated valuation of ecosystem services and trade-offs) software suite. The model uses monthly climate, topography, and land-use data to compute spatial indices of groundwater recharge, baseflow, and quickflow. We illustrate the model application in two large basins in Peru and Myanmar, where we compare results with observed data and alternative hydrologic models. We show that the spatial distribution of baseflow contributions correlated well with an established model in the Peruvian basin ($r^2 = 0.81$ at the parcel scale). In Myanmar, the model shows an overall satisfactory performance for representing month to month variation (Nash-Sutcliffe-Efficiency 0.6–0.8); however, errors are scale dependent highlighting limitations in representing processes in large basins. Our study highlights modeling challenges, in particular trade-offs between model complexity and accuracy, and illustrates the role that parsimonious models can play to support watershed management programs.

1. Introduction

Watershed management increasingly incorporates information on ecosystem services, the benefits that humans derive from nature (Guswa et al., 2014; Romulo et al., 2018; Vogl et al., 2017). Hydrologic ecosystem services comprise regulating flow to avoid flood damage, supplying water during the dry season, improving water quality, as well as a number of cultural – aesthetic, spiritual – services (Brauman, 2015). By implementing programs that protect or restore ecosystems, water resources managers can enhance hydrologic services that benefit downstream water users. Water quality and baseflow enhancement are often priorities for watershed management programs, which are frequently implemented in areas affected by agriculture (Bremer et al., 2016a, 2016b). Baseflow is defined here as the “streamflow fed from deep subsurface and delayed shallow subsurface storage between precipitation and/or snowmelt events” (Ward and Robinson, 1990). Protecting or enhancing baseflow is important for urban and rural water supply, particularly in the dry season and in developing nations where water-storage infrastructure may be more limited (Guswa et al., 2017). For example, about 60% of households in rural areas rely on groundwater for drinking in South-East Asia (Carrard et al., 2019).

One major challenge for program managers and scientists supporting watershed management programs is to understand where baseflow originates in the landscape and how this may change under future

* Corresponding author. 371 Serra Mall, Stanford, CA, 94305, USA.
E-mail addresses: perrine.hamel@ntu.edu.sg (P. Hamel), j.valencia@cgiar.org (J. Valencia), rschmitt@stanford.edu (R. Schmitt), manish.shrestha@sei.org (M. Shrestha), thanapon.piman@sei.org (T. Piman), rpsharp@stanford.edu (R.P. Sharp), w.francesconi@cgiar.org (W. Francesconi), aguswa@smith.edu (A.J. Guswa).
[1] Present address: Asian School of the Environment, Nanyang Technological University, 50 Nanyang Ave, 639798 Singapore.

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conditions, e.g. alternative land use and management practices. Spatially-explicit hydrologic models are one way to obtain information on baseflow processes in the landscape. However, the development of models capable of predicting baseflow generation in a fully-distributed manner remains a research frontier in hydrology (Brauman, 2015; Guswa et al., 2014; Hrachowitz et al., 2014, 2013). Existing spatially-distributed models often require parameterizing hydrologic processes for a large number of landscape units (e.g. SWAT, RHESSys, Guzman et al., 2015; Tague and Band, 2004), which necessitate extensive analyses to verify that the model is a valid representation of baseflow generation processes. Such verification analyses require observed data such as flow time series at multiple gauges to assess the spatial distribution of baseflow generation (Bennett et al., 2013), data that are lacking in many regions. In addition, a high number of model parameters makes these models more suitable for expert users and greatly increases the complexity of model calibration (Hrachowitz et al., 2013).

An alternative approach to characterize the baseflow supply across the landscape is to focus on its main contributing factors. These include climate, geology, and land use (Devito et al., 2005; Jencso and McGlynn, 2011; Price, 2011; Thompson et al., 2011b), whose relative importance depends on the hydrogeologic context. Topographic index models have been developed in hydrology based on the concept that topographic location and geological properties determine the likelihood of runoff generation (Agnew et al., 2006; Beven and Kirkby, 1979). However, most of these models focus on total streamflow and not specifically on baseflow, or remain conceptual for lack of quantitative relationships (Thompson et al., 2011b). The InVEST seasonal water yield (SWY) model used in this paper is an attempt to advance research in this direction and to quantify multiple aspects of the water budget on a landscape scale. Specifically, it combines spatial information on climate, topography, and land use with simple assumptions about flow routing, to provide spatially fully distributed insights into relative baseflow contribution within a landscape. Importantly, the model was developed for applications in data scarce environments, where the need for decision-support tools is great.

Although the InVEST-SWY model has been used in several studies globally, there is limited information on its hydrologic performance. Sahle et al. (2019) and Yang et al. (2018) have applied the model in Ethiopia and China, respectively, but did not compare model outputs to observations or alternative model predictions. Similarly, Wang et al. (2018) and Bagstad et al. (2018) have performed sensitivity analyses for watershed in Australia and Rwanda, respectively, but their analyses do not provide insights into the model's accuracy. To our knowledge, only Scordo et al. (2018) have compared InVEST-SWY model's outputs in an extensive study of 749 watersheds in Northern America. However, their study focused on the quickflow component of the model, ignoring the baseflow component of interest in our research.

To further examine the InVEST-SWY model's hydrologic performance, this study presents its application in two watersheds: Cañete, Peru, and Chindwin, Myanmar. These case studies are typical examples of watershed management programs being designed in data-scarce settings for which the model was developed. Despite data scarcity, previous studies and available data in each site allow validating the key outputs of the InVEST-SWY model. In the Cañete watershed, previous modeling efforts with another process-based hydrologic model, the soil water assessment tool (SWAT), provide a benchmark to assess the spatial distributions of various components of the water budget. For the Chindwin watershed, gauge records of discharge at different points in the catchment are available to test the modeled hydrologic response in space and time. In addition, the case studies represent different scales and climate types: the Cañete watershed is 6000 km^2, with an average of 1000 mm of annual precipitation, whereas the Chindwin basin spans across 114,000 km^2, with 3500 mm of annual precipitation.

The following sections briefly present the InVEST-SWY model and illustrate its application in the context of landscape management in our two case studies, each representing a different decision context. Since the InVEST-SWY model was developed as a practical tool to inform watershed management, the analyses aim to i) demonstrate how the model can be quickly set up for a new region; ii) assess the sensitivity of the model to flow routing parameters and characterize associated uncertainty; and iii) validate model outputs by comparison with observed data and alternative model outputs. We conclude with a discussion of the strengths and limitations of the InVEST-SWY model in terms of process representation and appropriate scales of application.

2. Materials and methods

2.1. The InVEST seasonal water yield model

The InVEST-SWY model was developed to provide a spatial estimate of baseflow production in a watershed. The model also provides monthly estimates of surface runoff, which explains its name in the software suite (seasonal water yield). Its conceptual framework is similar to a network water balance model developed by Thompson et al. (2011a,2011b), which presents two key features: sensitivity to vegetation (and hence land use), and explicit representation of routing (which allows for a GIS implementation). The model has low data requirements – with inputs drawn from globally available DEM (Digital Elevation Model), climate, and soil data – and relies on basic principles of water partitioning (precipitation becoming runoff or evapotranspiration) and routing (upgradient water becoming available to downgradient parcels). The simplified routing processes mean that model outputs have a large degree of uncertainty and should be interpreted as indicators of change rather than predictions of absolute values.

A full model description is provided in the user's manual (Sharp et al., 2019). Here, we present the main four computational steps and describe the variables and parameters used in our study, illustrated in Fig. 1. First, the model calculates the monthly quickflow (*QF*) on each pixel based on a modification to the NRCS curve number approach (NRCS-USDA, 2004), which estimates monthly direct runoff based on monthly precipitation and the number of rainfall events per period (Guswa et al., 2018).

Next, the model partitions the monthly available water between local recharge and evapotranspiration. On a given pixel, that partitioning is affected by upgradient recharge, and three parameters, α, β, and γ, govern the availability of subsurface water for evapotranspiration.

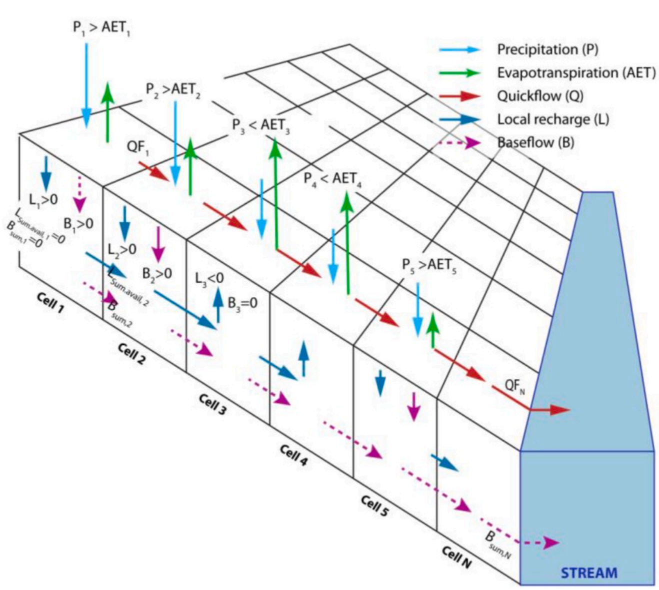

Fig. 1. Routing scheme used in the InVEST seasonal water yield model. A simplified (1D) flow path is represented to illustrate the simplified water balance calculated for each pixel.

Specifically, evapotranspiration $AET_{i,m}$ is computed as follows, expressing its limitation by either potential evapotranspiration ($PET_{i,m}$, the energy demand) or available water:

$$AET_{i,m} = \min(PET_{i,m}; P_{i,m} - QF_{i,m} + \alpha_m \beta L_{sum.avail,i}) \quad [1]$$

where $L_{sum.avail,i}$ is the sum of upgradient subsurface water that is potentially available at pixel i, β is a spatial availability parameter (0–1) that reduces the available water based on local topography and geology position (e.g., a concave position on the landscape versus a convex position), and α_m is the fraction of the (annual) upgradient subsidy that is available in month m (with a default value of 1/12). In this sense, β is analog to the parameter k_u in the conceptual model by Thompson et al. (2011a,2011b), which characterizes drainage: in the InVEST-SWY model, we added a parameter α_m to represent the temporal mismatch between evapotranspiration and subsidy (since not all annual subsidy may be available in month m). We note that the model has another optional parameter, γ, which determines the fraction of recharge at a given point that is available downgradient (i.e., not lost to deep groundwater storage). This parameter is set to 1 in the following analyses.

Next, the model computes the local recharge (L) on a pixel, which represents the *potential* contribution to baseflow, aggregated at an annual time scale:

$$L_i = P_i - QF_i - AET_i \quad [2]$$

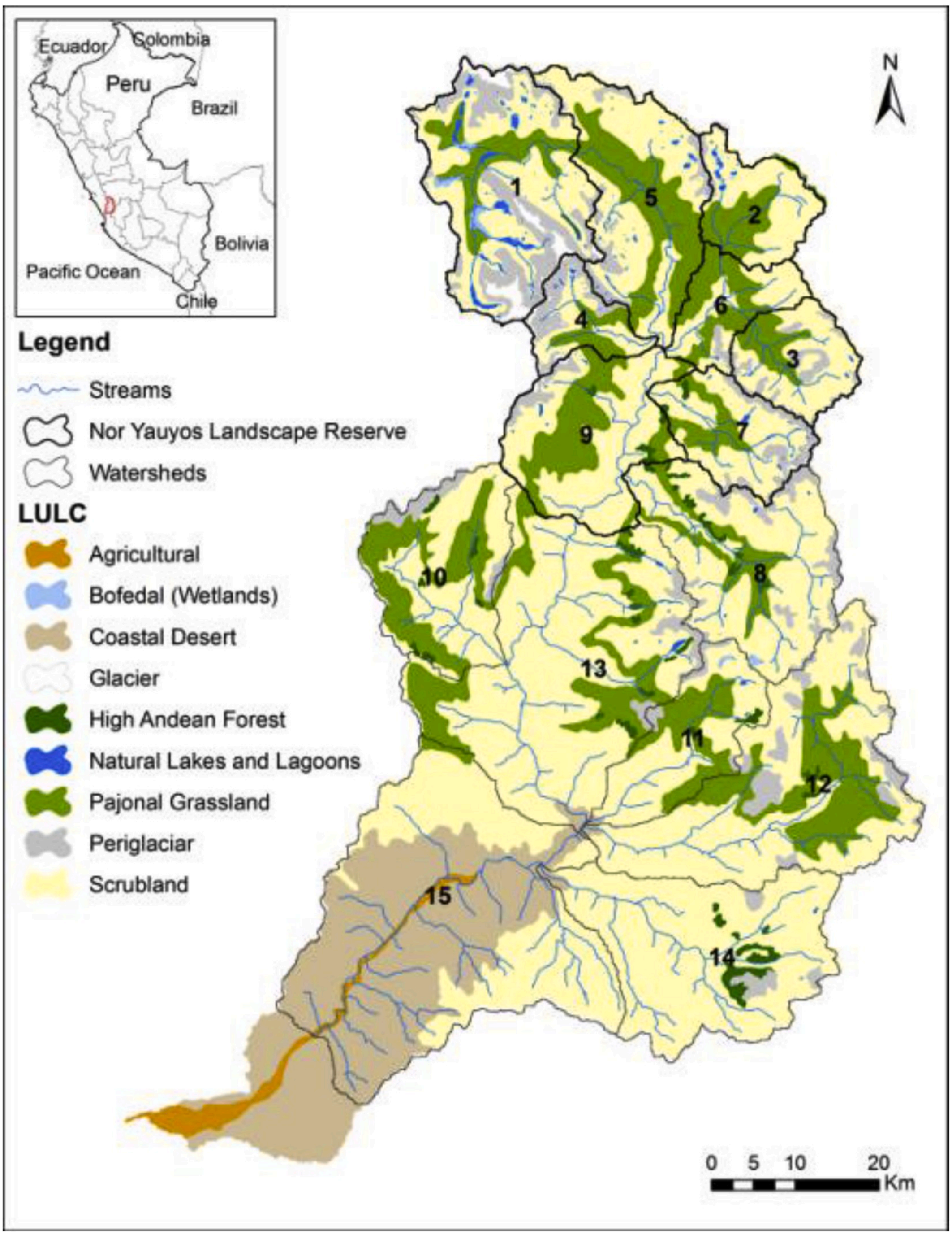

Fig. 2. Land use/ land cover and stream network in the Cañete basin. Numbers reference subwatersheds used in this study. The Nor Yauyos Landscape Reserve is located in the upper subwatersheds (delineated in black).

where P_i is annual precipitation, QF_i is the annual quickflow, and AET_i is the sum of monthly actual evapotranspiration, $AET_{i,m}$. We note that this simplified water balance ignores irrigation or groundwater withdrawals, terms that can be lumped into the precipitation input.

Finally, the model computes the baseflow index (B), which represents the actual contribution of a pixel to baseflow (i.e. water that reaches the stream). If the local recharge is negative, then the pixel did not contribute to baseflow so B is set to zero. If the pixel contributed to groundwater recharge, i.e. if L is greater than zero, then B is a function of the amount of flow leaving the pixel (B_{sum}, Fig. 1) and of the relative contribution to recharge of this pixel, which is calculated recursively (starting from the pixel adjacent to the stream, $B_{sum,N}$).

2.2. Description of the case studies

2.2.1. Cañete basin

The Cañete basin is located in the department of Lima in west central Peru (UTM coordinates 8′543,759 N–8′676,000 N and 345,250–444,750 E) (Francesconi et al., 2018). The basin comprises subtropical deserts in the lowlands, shrublands in the midlands, and pastures in the highlands as well as some glaciers in the upper areas (Fig. 2). The most extensive agricultural activity at the upper watershed includes livestock grazing of cattle, sheep and camelids. The Cañete basin covers an area of approximately 6192 km^2, from an elevation of around 4400 m to sea level. Precipitation at the basin changes drastically with elevation, from about 17 mm annually in the lower watershed to about 1000 mm in the upper areas. In the upper and middle areas, soils are predominantly alkaline and have low organic matter content and sandy loam and loam textures, common in paramo and puna ecosystems found in the Peruvian Andes. However, in the mountain areas where relic forests or prairie wetlands ("bofedales" in spanish) are found, soils are acidic with a higher degree of organic matter content and are characterized by silty clay textures. Recent concerns about water supply in the region have led a group of stakeholders – the Nor Yauyos Landscape Reserve (see Fig. 2), the Ministry of the Environment and the National Superintendence Sanitation Service – to consider landscape management options that would mitigate the risk of water scarcity. Given the impact of poor livestock management practice on soil properties and runoff, several studies were conducted to assess how improvements in grazing management could help mitigate future water scarcity (Francesconi et al., 2018; Hamel et al., 2019; Quintero et al., 2013).

In this case study, we use the InVEST-SWY tool as a spatially-explicit model of baseflow contribution to assess the effect of land-use policy on water resources. This information can be compared with spatial information on additional services (e.g. agricultural productivity) to develop optimal landscape management scenarios, as was done in a separate study (Hamel et al., 2019).

2.2.2. Model inputs for the Cañete basin

Climate and DEM inputs for InVEST-SWY were obtained from global and local sources described in Table 1. To the extent possible, for the purpose of model comparison, we used common data sources for the InVEST-SWY model and the SWAT model developed by Francesconi et al. (2018). For example, climate data were obtained from time series at eight weather stations (Francesconi et al., 2018) and aggregated at the monthly time step for the InVEST-SWY model. Details on SWAT inputs and the calibration process are provided in Appendix A.

2.2.3. Chindwin basin

The Chindwin basin is located in North West Myanmar (21^0 00′ N - 28^0 00′ N and 93^0 00′ E to 98^0 00' E), with a small portion in India (Fig. 3, inset). The basin covers an area of 114,000 km^2. The basin covers an elevation range from 50 to 3800 m. The mean precipitation is approximately 3500 mm annually, with most precipitation occurring during the monsoon from June to October. Around 15% of the basin is located in India, and the remainder (97,000 km^2) is located in Myanmar. There are three gauging stations in the Chindwin basin, namely Hkamti, Mawlaik, and Monywa. All three are located on the mainstem of the Chindwin and cover increasing drainage areas of 27,000 km^2 (Hkamti), 69,000 km^2 (Mawlaik) and 110,000 km^2 (Monywa). The Chindwin is one of the largest tributaries of the much larger Ayeyarwady River (Fig. 3, inset, blue), a river of major importance for livelihoods and biodiversity (Grill et al., 2019). The Chindwin is also subject to rapid land-use change especially because of the rampant expansion of illegal mining in the Northern Part of the basin and agriculture development in the downstream area of the basin. Water resources in the Chindwin are under pressure from development and climate change. An extreme drought event in 2010 caused the depths of the Chindwin River during the dry season to drop to levels much lower than those seen in the past. Low water levels resulted in water shortages for agricultures, domestic use and navigation. Piman et al. (2019) indicates that improving groundwater recharge may be an effective strategy to improve resilience under future conditions where such droughts might be more likely.

In this case study, we thus use the InVEST-SWY tool as a simple rainfall-runoff model to capture the monthly water budget at different locations in the basin. Based on available observations, we study not only how well InVEST-SWY is able to reproduce observed mean monthly stream flows, but also the partitioning into mean monthly baseflow and quickflow. The later experiment is relevant to understand if indicators of water balance components produced by InVEST-SWY (namely annual B and monthly QF) can be interpreted in absolute number or rather as relative values only. From a management perspective, a sound representation of stream flows can be of interest to judge changing magnitude and timing of high or low flows and related surface water availability or hydrologic hazards.

Table 1
SWAT and InVEST model inputs for the Cañete case study.

Name	Source	Note
LULC	National Geographic Institute (IGN)	Scale 1:100,000 Realized between 1970 and 1971
Soils (including soil group)	National Office of Natural Resource Evaluation (ONERN)	Scale 1:100,000 Realized in 1989
DEM	CGIAR-CSI SRTM (http://www.cgiar-csi.org/data/srtm-90m-digital-elevation-database-v4-1)	90 m of spatial resolution
Stream network, urban centers, snow-covered mountain peaks[b]	Nor Yauyos-Cocha Landscape Reserve, International Center for Tropical Agriculture (CIAT)	Snow melt only affects the upper watersheds [# 1–6 and #9] and is added to precipitation
Weather data	National Meteorology and Hydrology Service of Peru (SENAMHI) for eight weather stations (Francesconi et al., 2018) Interpolated (inverse distance weighted) for InVEST	Precipitation, maximum and minimum temperature, solar radiation, relative humidity and stream flows. For the period 1991–2009
Biophysical table[a]	CN values from SWAT report (Uribe et al., 2013) Kc[a] values from FAO (FAO, 1996) for the current landscape	
Rain events table	From IWMI (International Water Management Institute) Water atlas (http://wcatlas.iwmi.org/)	Coordinates: −75.8; −12.5

[a] InVEST inputs.
[b] SWAT only inputs.

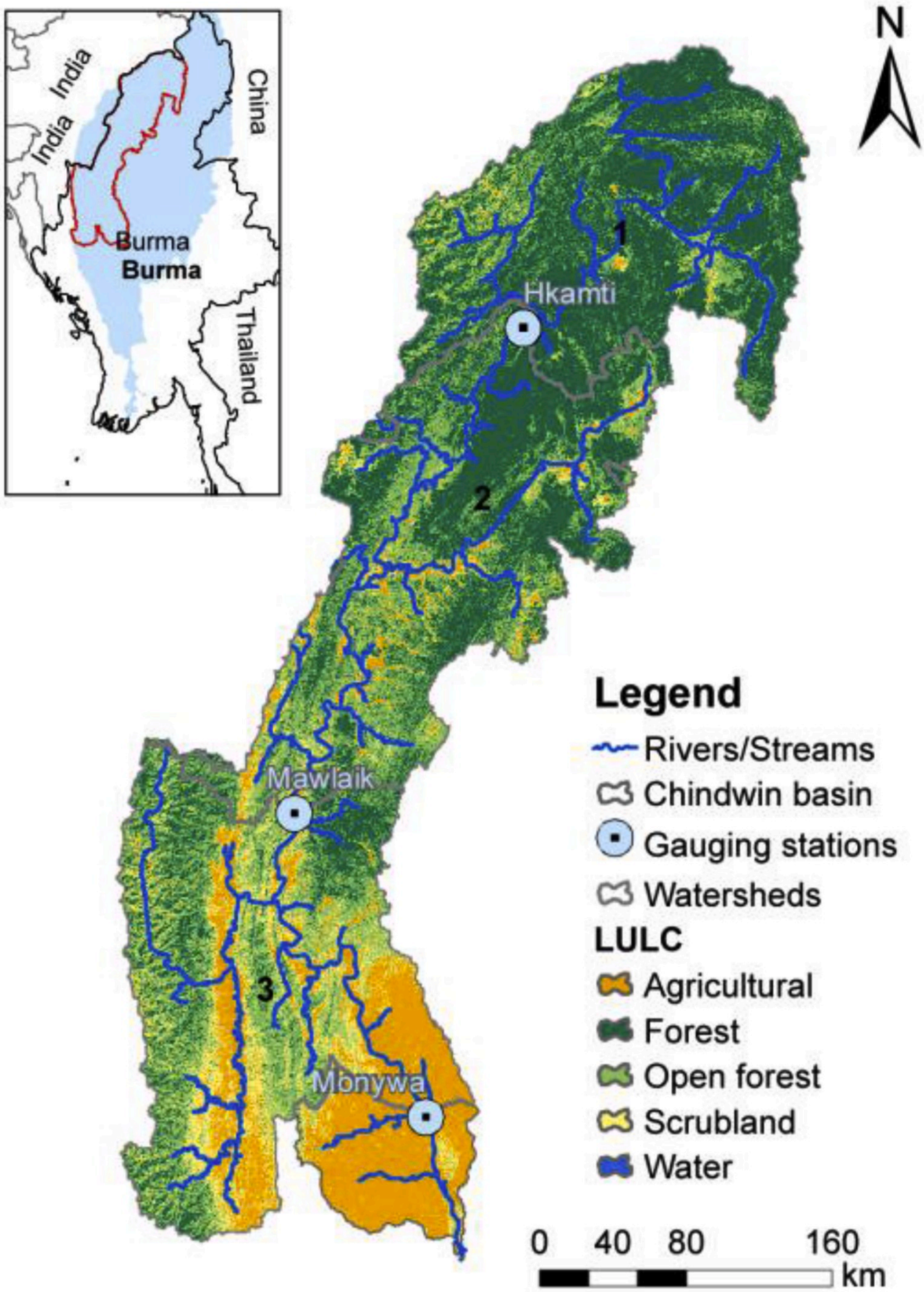

Fig. 3. Land use/ land cover and stream network in the Chindwin Basin, Myanmar. Markers show the location of gauging stations used in this study. Numbers reference subwatersheds used in this study.

2.2.4. Model inputs for the Chindwin basin

Climate and DEM inputs for InVEST-SWY were obtained from global and local sources described in Table 2. For the Chindwin case, precipitation and temperature data were generated using SimCLIM model (see Yin et al. (2013) for more information about SimCLIM model). As some data were obtained from the Mandle et al. (2017), which focused on Myanmar's territory, this analysis does not include parts of the basin located in India.

2.3. Sensitivity analyses and model calibration

To better understand the effect of the availability parameters (α and β), we conducted one-at-a-time sensitivity analyses for both parameters (Pianosi et al., 2016). First, with the default value for α (1/12), we varied β from 0 to 1 by increments of 0.2. A value of 0 for β means that there is no upslope contribution (see Fig. 1), while a value of 1 means that the entire contribution from upslope pixels is available for evapotranspiration downgradient. To assess the effect of α, we repeated the above analyses for values of α equal to 0, 1/6, and 1/3 (for all pixels). A value of 0 for α means that local recharge from upgradient is not available for evapotranspiration locally, representing the same case as $\beta = 0$ above. A value of 1/3 for α means that one third of the annual recharge, i.e. the equivalent of four average months, is potentially available downstream on a given month. This might occur in a

Table 2
InVEST model inputs for Chindwin case study.

Name	Source	Note
LULC	Custom map made by WWF from Google Earth Engine	Resolution: 150m; date: between 2013 and 2014
Soils (including soil group)	Harmonized World Soil Database	Resolution: 30 arc-second; date: 2012
DEM	CGIAR-CSI SRTM (http://www.cgiar-csi.org/data/srtm-90m-digital-elevation-database-v4-1)	Resolution: 90 m
Precipitation data	SimCLIM data set	For the period 1980–2005
Evapotranspiration	Calculated using Thornthwaite method. Temperature data was generated from SimCLIM data set	For the period 1980–2005
Biophysical table	Kc and CN values from national ecosystem services assessment (Mandle et al., 2017)	–
Rain events table	From national ecosystem services assessment (Mandle et al., 2017)	–

low-gradient watershed with slow release of recharged water. Of note, setting α such that its sum over a year is greater than one may violate the water balance (equation 1). In practice, we address this issue by verifying that the model outputs satisfied the annual water balance, i.e. that the extra water used in dry months was less than the total volume of recharge. These verifications were done at the subwatershed scale.

In the Chindwin basin, we also studied impacts of changing curve numbers on the representation of quickflow and baseflow. This experiment aimed to provide guidance on the application of the curve number approach which drives hydrologic partitioning in the InVEST-SWY model (see Section 2.1) at the monthly timescale. Specifically, we examined the effect of antecedent moisture conditions (AMC), which capture the soil saturation effect, on quickflow (NRCS-USDA, 2004). Additional details are provided in Appendix B. We did not conduct these analyses for the Cañete basin due to the difference in performance metrics: since the InVEST-SWY model is compared to the SWAT model in the Cañete basin, we used the curve number values set in SWAT for both models.

2.4. Model performance

Model comparison with SWAT. By comparing the InVEST-SWY model with a calibrated SWAT model in the Cañete basin (Appendix A), we aim to study how well the model reproduces the spatial distribution of hydrologic indicators at the basin scale. Model comparison is recognized as a robust method to produce insights given the uncertainties in ecohydrological processes (Bagstad et al., 2018; Sivapalan and Blöschl, 2017; Vigerstol and Aukema, 2011). From a management perspective, a sound representation of spatial variability helps prioritize areas for intervention. Our variables of interest are the groundwater recharge and baseflow contribution, as well as the surface runoff (quickflow). In SWAT, groundwater recharge is represented by the output GW_RCHRG, which, over long time periods, is equivalent to percolation (PERC). Groundwater recharge encompasses recharge to both the deep and shallow aquifers. Baseflow contribution is calculated as:

$$\mathrm{LATQ} + \mathrm{GWQ} - \mathrm{TLOSS} + \mathrm{SW\_end} - \mathrm{SW\_init}, \quad [3]$$

where LATQ is the lateral flow, GWQ is the groundwater contribution, TLOSS is the transmission loss in the stream channel, SW_end and SW_init are the soil water storage at the end and start of the simulation, respectively.

In the InVEST-SWY model, groundwater recharge and baseflow contribution are represented by the L and B outputs, respectively (see Section 2.1). Because snowmelt is not represented in InVEST, we add that component (derived from SWAT) to B values.

For our variables of interest, we compared output rasters from InVEST and SWAT by computing correlation coefficients at the subwatershed scale (n = 15). To evaluate results relative to the annual water balance, we also compared the subwatershed surface runoff and annual evapotranspiration from both models.

Model comparison with observed data. In the Chindwin basin, we compared the InVEST model with observed discharge data at different locations within the basin, each with different drainage areas. This comparison was designed to estimate how well InVEST-SWY reproduces different components of the monthly water balance spatially and temporally.

We used 20 years of observed daily discharge data at three stations in the Chindwin Basin, which we converted to water yield (from m^3/s to mm/d). We then separated the observed daily hydrographs into a baseflow and a quickflow component using an automated baseflow separation technique (Arnold and Allen, 1999) for each of the three time-series. We then aggregated the daily values into mean monthly quickflow and baseflow values over the 20-year observation period. The observed quickflow, baseflow, and total water yield are denoted as QF_{obs}, BF_{obs}, and $Qtot_{obs}$.

Since InVEST-SWY does not directly produce streamflow values (but rather annual baseflow and monthly quickflow), we estimated monthly streamflow at each station as the sum of monthly quickflow, QF, and monthly baseflow B from upstream pixels. Given that observed baseflow values showed strong seasonality, we distributed monthly baseflow values across the year following the same distribution as relative precipitation (monthly values relative to total values), shifted by one month. It should be noted that this calculation of monthly streamflow assumes an instantaneous arrival of quickflow to the stream, i.e. that quickflow generated during month m reaches the downstream gauge in that same month.

We compared InVEST-SWY model results with observed streamflow and the quickflow component extracted from the observed hydrograph. First, to assess the performance of the model spatially, we compared modeled and observed baseflows for the three stations. Next, we calculated the Nash-Sutcliffe Efficiency for quickflow and total streamflow as:

$$NSE_g = 1 - \frac{\sum_{i=1}^{12} \left(X_{obs}(g,i) - X(g,i)\right)^2}{\sum_{i=1}^{12} \left(X_{obs}(g,i) - \overline{X_{obs}(g,i)}\right)^2} \quad [4]$$

where X is QF or Q, for quickflow and total flow, respectively.

Last, we calculate the monthly residual for quickflow and total discharge as:

$$\varepsilon_X(g,m) = X(g,m) - X_{obs}(g,m) \quad [5]$$

where X is QF or Q, for quickflow and total flow, respectively. We also estimate the total cumulative error in quickflow and discharge by summing the residuals over the months.

3. Results

3.1. Sensitivity analyses

The sensitivity analyses highlighted the distinct roles of the flow routing parameters α and β. In all cases, larger values of α and β indicate the potential for more recharge to be removed via evapotranspiration and concomitantly to reduce baseflow. In both basins, we found that modeled values of B were much more sensitive to β (Fig. 4, left panels) than to α (Fig. 4, right panels). The left panels in Fig. 4 show that the relationship between subwatershed values of B and β is concave, with a drop as high as 140 mm (subwatershed #6) or as little as 0.01 mm (subwatershed #15) between extreme values of β (0 and 1) in the Cañete basin. The sensitivity of B to β determined with other values of α (1/6

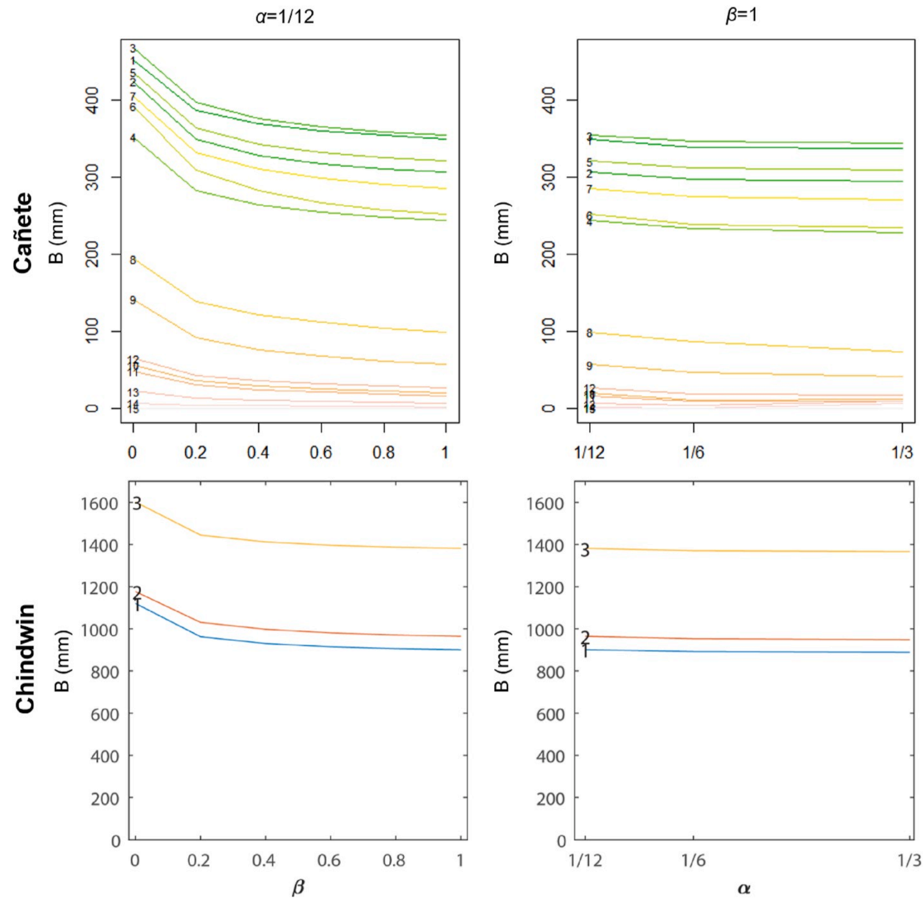

Fig. 4. One-at-a-time sensitivity analysis of the InVEST-SWY model baseflow, B, to α (right) and β (left) parameters. Top row shows results for Cañete, while bottom row is for Chindwin. B values are aggregated per subwatershed. Each line represents a subwatershed, identified by its number on the left-hand side (as shown in Fig. 2 and 3). Note the difference of scale for the y-axis.

and 1/3) yielded similar results, although the maximum drops in values were slightly higher (153 mm and 158 mm, respectively, for subwatershed #6). In the Chindwin basin, varying β had the greatest impact on B for the most downstream station (#3), Monywa, where B varied between 1600 and 1380 for $\beta = 0$ and $\beta = 1$. The relative change was relatively similar for all three catchments and ranged from 20% to 14%.

The parameter α governs the temporal availability of upslope subsidies. With the interval used in these analyses, α had a lower effect on the baseflow contribution B (Fig. 4, right panels). Varying α between extreme values (1/12 to 1/3) affected baseflow by less than less than 2% for the Chindwin subwatersheds and by less than 7% for the Cañete subwatersheds with high water yield (>200 mm, subwatersheds #1–7). For dry subwatersheds, the relative change was larger (>25%) but this corresponded to small changes in baseflow (<25 mm).

The model also showed a high sensitivity to CN values. Quickflow values increased by factors of 10.1, 12.3, and 13.1, respectively, for the Hkamti, Mawlaik, and Monywa watersheds in the Chindwin basin (392 mm, 392 mm, and 424 mm, respectively, see Appendix B). Partitioning between quickflow and baseflow was closer to observations with wet AMC values. This is expected given climate conditions in a subtropical environment where most precipitation falls during a well-defined rainy season. Thus we used wet AMC in the following analyses.

3.2. Model performance

3.2.1. Comparison with SWAT

In the Cañete basin, the InVEST-SWY model showed high baseflow contribution from the upper eight subwatersheds where precipitation is the highest (Fig. 5). Baseflow estimates ranged from 0 to 1354 mm, a range equal to that of groundwater recharge estimates. The baseflow estimates from SWAT showed similar patterns to InVEST throughout the basin (Fig. 5), although the range was larger (from 0 to 1597 mm). For groundwater recharge, the order of magnitudes predicted by the two models were consistent (InVEST-SWY estimates ranging from 0 to 1354 mm, whereas SWAT estimates ranged from 0 to 1142 mm).

At the subwatershed scale, the statistics reflected a general agreement between the two models of baseflow estimates (Fig. 6), with a r^2 value of 0.98. At the pixel scale, there was higher discrepancy between the two models due to the aggregation of values by HRU in SWAT. Although there was a strong correlation ($r^2 = 0.81$), the root mean square error was high (208 mm).

Surface runoff estimated from SWAT was higher than InVEST-SWY's estimate, reaching 110 mm (9% of the precipitation) vs. 43 mm (less than 5% of the precipitation) for InVEST-SWY. Evapotranspiration in SWAT was 36% of precipitation for the upper subwatersheds and reaching 86% for the lower subwatersheds. In InVEST-SWY, evapotranspiration ranged from 46% (upper subwatersheds) to 100% of precipitation (lower subwatersheds). A baseflow analysis (Lyne and Hollick filter, implemented in the R package 'hydrostats' with coefficient a = 0.975) indicated that the annual baseflow index averages 0.68, suggesting that surface runoff contributes less than 32% to the total streamflow.

3.2.2. Comparison with observed data

The comparison with observed data in the Chindwin basin showed mixed results. From observed streamflow data, annual runoff is 1531 mm/yr for Hkamti, 1198 mm/yr for Mawlaik, and 877 mm/yr for Monywa (Table 3). InVEST-SWY did not predict average streamflow

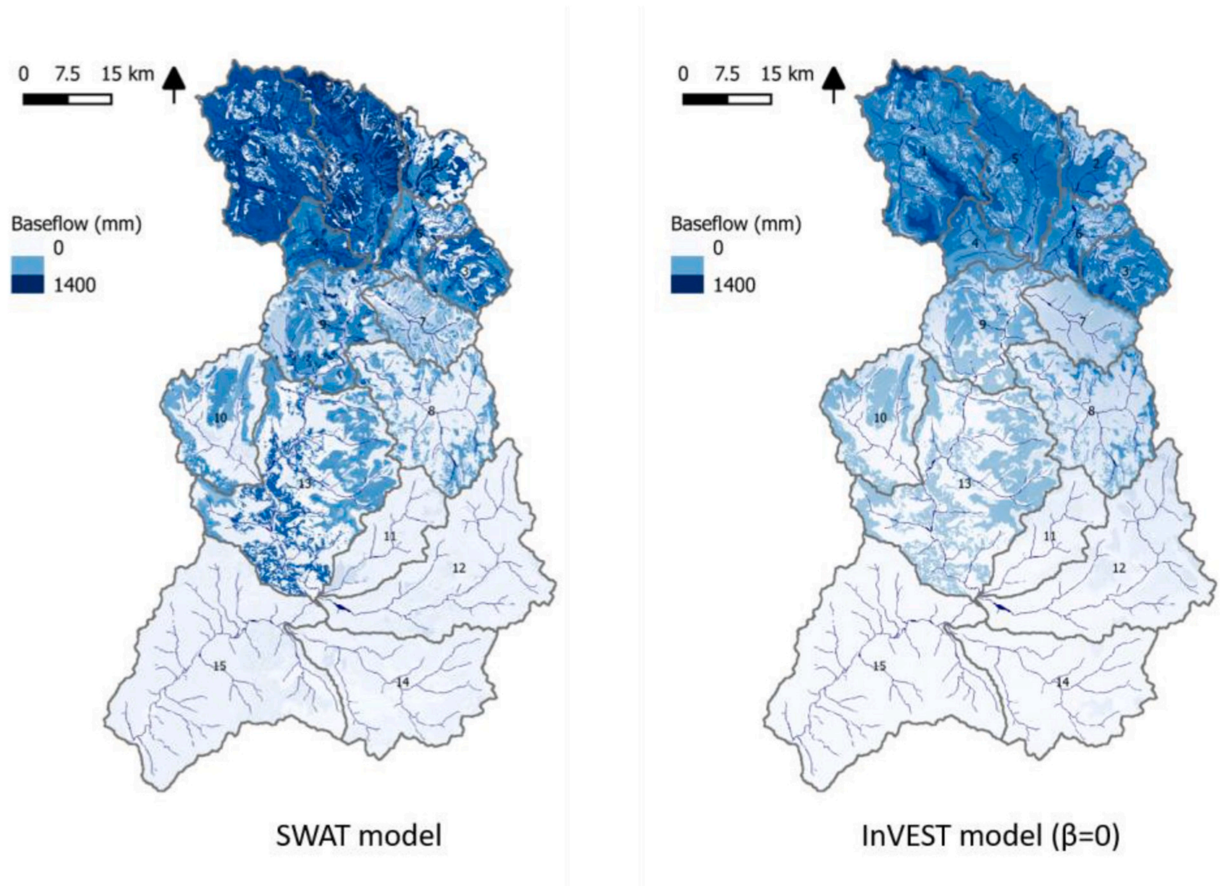

Fig. 5. Comparison of the InVEST-SWY and SWAT baseflow estimates.

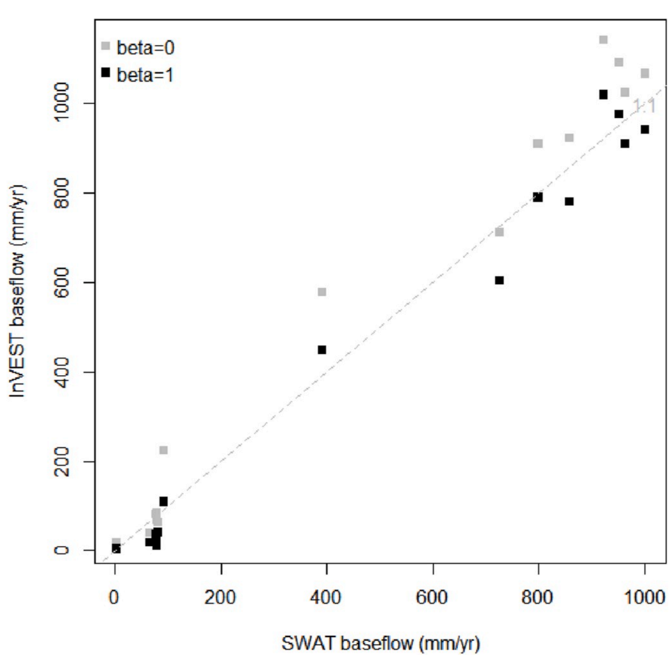

Fig. 6. Comparison between InVEST-SWY and SWAT models for baseflow aggregated for each of the 15 subwatersheds. InVEST-SWY baseflow estimates are given for $\beta = 0$ and $\beta = 1$.

Table 3
Summary of annual InVEST values, using wet antecedent moisture conditions, and observations at the three stations. See stations on Fig. 3.

All flows in (mm/yr)	Hkamti	Mawlaik	Monywa
InVEST-SWY total flow	954	986	1139
InVEST-SWY baseflow (and %total flow)	524 (55%)	557 (56%)	682 (60%)
Observed total flow	1531	1198	877
Observed baseflow (and %total flow)	745 (49%)	689 (58%)	551 (63%)

well, possibly because of the large interannual variability (as shown in Fig. 7 for monthly values) and to errors in curve number or crop coefficient values (both associated with the land use/ land cover). Total flow is sometimes overpredicted (Monywa) sometimes underpredicted (other two stations), which could be related to the land use response (the larger watershed, Monywa, comprises much more agriculture and open forest than the upper subwatersheds (Fig. 3)). The sign change in the residuals for the downstream watershed could also be linked to a spatial scale in the error processes, e.g., stream routing not being explicitly considered in the model.

Partitioning between quickflow and baseflow estimated from daily time series shows that at the annual time scale, the proportion of baseflow relative to total flow, or baseflow index, is well represented by InVEST-SWY: the baseflow index values are within 6% of observed values (Table 3). However, while the baseflow component decreases in absolute values for more downstream stations, that pattern is not represented in the InVEST-SWY results. Rather, the proportion of baseflow relative to total flow is similar for all stations (Table 3, see Discussion).

Fig. 7 presents the modeled streamflow values to observations for the three stations, showing that modeled monthly values are generally within the range of ± one standard deviation of the observations. The

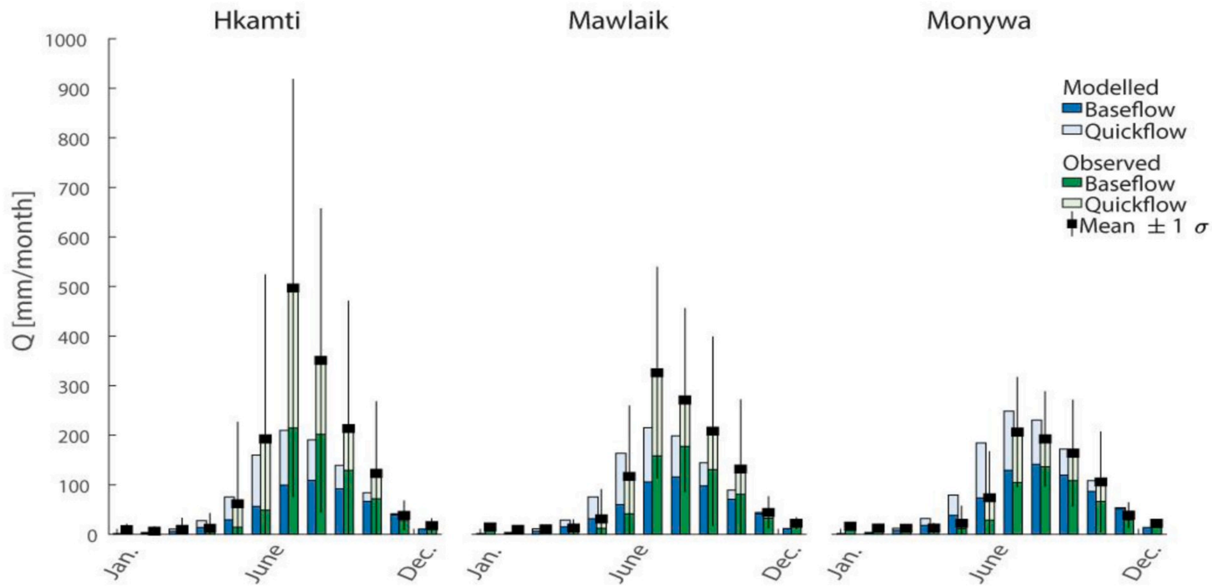

Fig. 7. Comparison between InVEST-SWY model predictions and monthly observations at three stations in the Chindwin Basin (Fig. 3). Hydrographs represent modeled and observed quickflow and baseflow.

monthly distribution of baseflow according to precipitation distribution (Fig. 7) seemed to match observations. Comparing modeled and observed streamflow for all months shows a good fit between modeled and observed values (NSE of 0.58, 0.8, and 0.68 respectively, for Hkamti, Mawlaik, and Monywa stations).

4. Discussion

The analyses presented in this paper provide insight into the strengths and limitations of the InVEST-SWY model for rapid assessment of quickflow and baseflow contributions. In the following, we discuss the interpretation of model outputs, the comparison analyses with the SWAT model, and the practical implications of this work.

4.1. InVEST seasonal water yield model parameters and sensitivity analyses

The InVEST-SWY model is parsimonious, with three availability parameters and two biophysical parameters (K_c and CN) per land use/ land cover. Used with globally available data for monthly climate, as well as topography, geology, and land use data, the model presents the advantage of being rapidly set up in a new geography. The model relies on a water balance at the pixel scale and the assumption that upslope subsidies are partially available to a given pixel, depending on its location on the landscape and upslope geological and land use characteristics. In that way, it links local scale (pixel) and hillslope scale, addressing an important challenge in ecohydrological models (Brooks et al., 2015; Moore et al., 2015; Sivapalan and Blöschl, 2017). In its parsimony, the InVEST-SWY is similar to the model developed by Batelaan and De Smedt (2007), with the difference that lateral water movement is made explicit in InVEST-SWY (as opposed to its integration with a groundwater model).

This explicit, simplified flow routing is one of the main disadvantages of the model since it is introduces significant uncertainty: for each pixel, closing the water balance at the monthly time step requires knowledge of upslope subsidies (to calculate available water for evapotranspiration). However, this knowledge is not readily available nor measurable for a given watershed (Brooks et al., 2015; Moore et al., 2015; Thompson et al., 2011b), making it difficult to select the values of α and β.

Uncertainty in flow routing can be assessed by sensitivity analyses such as the ones presented in this study. We showed that the β parameter, governing flow routing, had an important effect on the baseflow and groundwater recharge indices, with the baseflow *B* output varying by almost 50% for some subwatersheds. In a study of Northern American watersheds, subsurface flows at the hillslope scale were found to increase evapotranspiration by at least 20% (Thompson et al., 2011b, 2011a), significantly affecting the amount of water available in the stream. In theory, uncertainty about flow routing depends on climate and will decrease in wetter regions, when water deficit for evapotranspiration is low: in the extreme situation where precipitation is higher than evapotranspiration at all times, the local water balance will not be influenced by the water surplus coming from upslope areas. Our results seem to confirm this relationship since the model in two humid climates showed little sensitivity to α (Fig. 4).

Alternatively, uncertainty in flow routing could be assessed, and even reduced, by comparison with spatially explicit models or even observed data (streamflow or actual evapotranspiration), which is the approach illustrated in this study using two different case studies: one comparing the model with SWAT, the other with observed data. Of note, the hierarchy of drivers of baseflow generation (climate, land use, and topography) is still a topic of active research (Devito et al., 2005; Patil et al., 2012; Thompson et al., 2011b) limiting the reliability of many hydrological models (which use rules for the partitioning and fate of surface and subsurface flow). Simple models such as InVEST may be useful tools to test the effect of flow routing assumptions in a given climate and produce relevant baseflow information.

4.2. Model performance

In the Cañete basin, the estimate of subwatershed baseflow obtained with the SWAT and InVEST-SWY models was very similar, both in terms of its magnitude (e.g. relative to quickflow and evapotranspiration) and its spatial distribution. This suggests that both models are sensitive to the key drivers of baseflow, in particular climate, since precipitation was extremely variable across the basin (from <5 mm to >1000 mm). This finding confirms the results from Wang et al. (2018) who found that the InVEST-SWY model was highly sensitive to climate inputs. Adding the snowmelt component was critical (an earlier version of the model without the snowmelt correction had a lower performance), confirming the recommendations from Scordo et al. (2018) to incorporate this component of the water balance for basins receiving snowfall. Finally, we note that the model could be further calibrated by changing the curve number values, thereby increasing the quickflow values closer to those from SWAT. Overall, the comparison between InVEST-SWY and SWAT, a model commonly used for water resources management (Vigerstol and Aukema, 2011), at the subwatershed scale suggests that the tool is adequate for scenario analyses: in particular, the model can be used to estimate changes in land use or in climate, as required in watershed management programs.

Despite a strong correlation, discrepancies between models at the pixel scale remain important (RMSE = 208 mm) and may be explained by two differences in model structure. First, while surface runoff production is obtained in both models with the SCS Curve Number method (although at a different time step – monthly in InVEST-SWY, daily in SWAT), subsurface flow routing differs. The InVEST-SWY algorithm relies on a simple water balance governed by the upslope subsidy (parameterized with α and β), whereas the flow routing in SWAT is limited by the spatial configuration of HRUs. This characteristic of the SWAT model makes the comparison of the two models challenging at the fine spatial scale (HRUs, which have a median area of 0.3 km^2), which is why we preferred the subwatershed scale. Alternative models with explicit surface and subsurface flow routing, like RHESSys (Tague and Band, 2004), may be more appropriate for comparison with InVEST-SWY. Second, the precipitation interpolation in InVEST-SWY differs from SWAT: to create the precipitation raster for InVEST-SWY, we used inverse-distance weighting, which has shown good results for precipitation interpolation. However, in SWAT, precipitation was aggregated at the subwatershed scale, smoothing the differences within a subwatershed.

In Chindwin, the model was assessed on its ability to represent monthly water balance, baseflow contribution relative to total flow, and total streamflow. We found large errors in streamflow estimates (up to 38% for the upstream watershed) but the ability of the model to estimate baseflow contribution relative to total streamflow, assuming that baseflow values were distributed according to monthly precipitation, was fair. Performance of the InVEST-SWY reached a Nash-Sutcliffe Efficiency between (NSE) between 0.6 and 0.8, a reasonable performance for a rapid assessment tool built on global data. Recently, a lumped conceptual hydrologic model, MIKE NAM, was set up with detailed data to simulate flows at mainstream stations in Chindwin (Shrestha et al., 2020). The calibration and validation results show that the NSE of model performance varies between 0.78 and 0.86, comparable with InVEST (although MIKE NAM was assessed on daily, not monthly, data). Model performance was found to be scale-dependent and to vary between different stations, with positive residuals for more upstream stations and negative residuals for downstream stations. This could be related to heterogeneity in the catchment not being represented in the input data, or to the flow routing which is performed as instantaneous sum of upstream pixels rather than via an explicit open-channel scheme.

As noted by others (Wang et al., 2018; Bagstad et al., 2018), the model is also very sensitive to land use and climate inputs and our study did not investigate their effect. In addition, the model did not predict the decrease in (absolute, per area) baseflow with increasing subwatershed areas. This could be due to the lack of sensitivity of the flow routing parameters to slope and soils, which could be changed in the model structure. The available datasets do not allow us to investigate this point further, though. In fact, since "observed" baseflow is determined by an automated separation technique, it may also be influenced by flood-peak attenuation occurring in the channel, which may influence the baseflow signal. In addition, antecedent moisture conditions have a great impact on model performance (Figure A1), calling for careful consideration on how to set curve number values in geographies with more complex transitions between dry and wet conditions.

In summary, the analyses presented in this manuscript provide useful insights for the application of the InVEST-SWY in two large basins. We highlight below main implications for the two case studies, to guide future research in other basins:

- Using the model to represent spatial variations in baseflow within the Cañete basin (6000 km^2) seems appropriate, after correction for snowmelt, as it compared well with the more established SWAT model; this suggests that the model can be used for scenario analyses at the subwatershed scale, although further tests should be conducted to assess the model response to land use and climate change in more details.
- Using the model to estimate the overall water balance (partitioning between evapotranspiration and recharge) in the large Chindwin basin (114,000 km^2) seems appropriate as the model compared fairly with observations. However, the spatial variations were not well captured, which may be due to the very large scale of the basin (implying that separate calibration procedures should be conducted for the three sub-basins examined in this study), and to uncertainties in climate and land use inputs that were not investigated in this study.
- These mixed results suggest that the model should be used with caution —comparing it with observed data or alternative models—in new geographies, especially when using the absolute values of baseflow contributions; however, after calibration, the model can provide a valuable rapid assessment tool for different components of the hydrologic cycle, and their sensitivity to land and climate change. Modeled relative baseflow contributions and results in smaller basins (<10,000 km^2) seem to be better supported by our data.
- Future research could further examine the behavior of the InVEST-SWY model by applying similar analyses in data-rich basins. In particular, additional comparisons with spatially-explicit models could shed light on the strengths and limitations of the routing process, ultimately allowing to inform model parameterization in data-poor environments.

5. Conclusions

This study illustrates the application of the InVEST-SWY model to support landscape management planning decisions. The model presents the advantage of being parsimonious and has low data requirements. It produces two indices representing local recharge and baseflow contribution. We illustrated the application of InVEST as a practical tool in two basins, in Peru and Myanmar. Our analyses suggest that the model structure, relying on basic hydrologic principles, is useful for basin-scale baseflow assessment and high-level month-to-month assessments of basin water budgets. The main limitation is the difficulty to validate the tool, both for static mapping (e.g. current landscape) and for land-use change or land-management simulations, such as the one presented in our case study. Promising research directions include further testing of InVEST with remotely-sensed data (e.g. actual evapotranspiration) or with physically-based models that provide robust spatial-explicit estimates of baseflow generation on the same spatial scale than InVEST.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Perrine Hamel: Conceptualization, Methodology, Formal analysis, Writing - original draft. **Jefferson Valencia:** Methodology, Writing - original draft, Visualization. **Rafael Schmitt:** Methodology, Formal analysis, Writing - original draft, Visualization. **Manish Shrestha:** Data curation, Writing - review & editing. **Thanapon Piman:** Data curation, Writing - review & editing. **Richard P. Sharp:** Software. **Wendy Francesconi:** Methodology, Writing - review & editing. **Andrew J. Guswa:** Conceptualization, Methodology, Writing - original draft.

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Appendix

A. SWAT model parameterization for the Cañete

SWAT is a commonly-used spatially-explicit, process-based model, operating at the daily (or sub-daily) time step. One particularity of the model is the discretization of the model in hydrological response units (HRUs), which are areas in the landscape with unique combinations of climate, slope, and land use. A complete model description can be found in the technical documentation (Neitsch et al., 2011). The SWAT model of the Cañete basin was developed in a previous study conducted by the International Center for Tropical Agriculture (CIAT), as part of an initial hydrological assessment of the basin to support the design of the reward for environmental services program (Uribe et al., 2013). Model inputs described in Table 2 were used, where possible, to parameterize the InVEST model.

The Cañete basin was divided into 15 subwatersheds and a total of 2926 HRUs (with areas ranging from 0.008 km^2 to 243.7 km^2) were delimited. A previous global sensitivity analysis using the one-at-a-time method in SWAT2005 identified the 26 most important parameters for streamflow estimation. From this analysis, a total of 20 parameters were modified manually (Uribe et al., 2013). The most sensitive parameter was the Temperature Lapse Rate (TLAPS), which estimates the changes in temperature using elevation bands. Elevation bands help incorporate precipitation and temperature variation in watersheds with drastic changes in elevation (Stehr et al., 2009). Other important calibration parameters related to runoff estimation, such as the initial curve number value under an average soil moisture condition (CN2) and surface runoff lag time (SURLAG), as well as aquifer properties, groundwater processes, and snow melt, since the Cañete is a glacier-fed basin. The performance of the model was assessed with the use of the Nash-Sutcliffe Efficiency (E_{NS}) and the coefficient of determination (R^2). The SWAT model in the Cañete basin showed performance metrics of $R^2 = 0.62$ and $E_{NS} = 0.3$ at daily time step and $R^2 = 0.92$ and $E_{NS} = 0.82$ at monthly time step, suggesting that the model was capable of predicting water yields for the basin (Uribe et al., 2013).

B. Sensitivity analyses on curve number for the Chindwin basin

We tested the sensitivity of the model to Curve Number values for all land use types (Figure A1). Default CN values for a land use represent average soil moisture conditions. For wetter or drier conditions, $CN_{II}(LU)$ should be modified according to

$$CN_I = \frac{4.1CN_{II}(LU)}{10 - 0.058CN_{II}(LU)} \quad [a1]$$

And

$$CN_{III}(LU) = \frac{23CN_{II}(LU)}{10 + 0.13CN_{II}} \quad [a2]$$

To represent different runoff behaviors under very dry and very wet conditions. Where $CN_I(LU)$ should be used if there was less than 35 mm of precipitation in the previous 5 days and $CN_{III}(LU)$ should be used if there was more than 53 mm of precipitation in the previous 5. However, as InVEST-SWY operates on a monthly timescale and with a statistical distribution of runoff events, this correction is not easily applicable in the model, Thus, we tested the model using both CN_{II} and CN_{III} conditions. We did not test the CN_I case, which is not very relevant in a sub-tropical climate when most precipitation and runoff will occur during a well-defined rainy season.

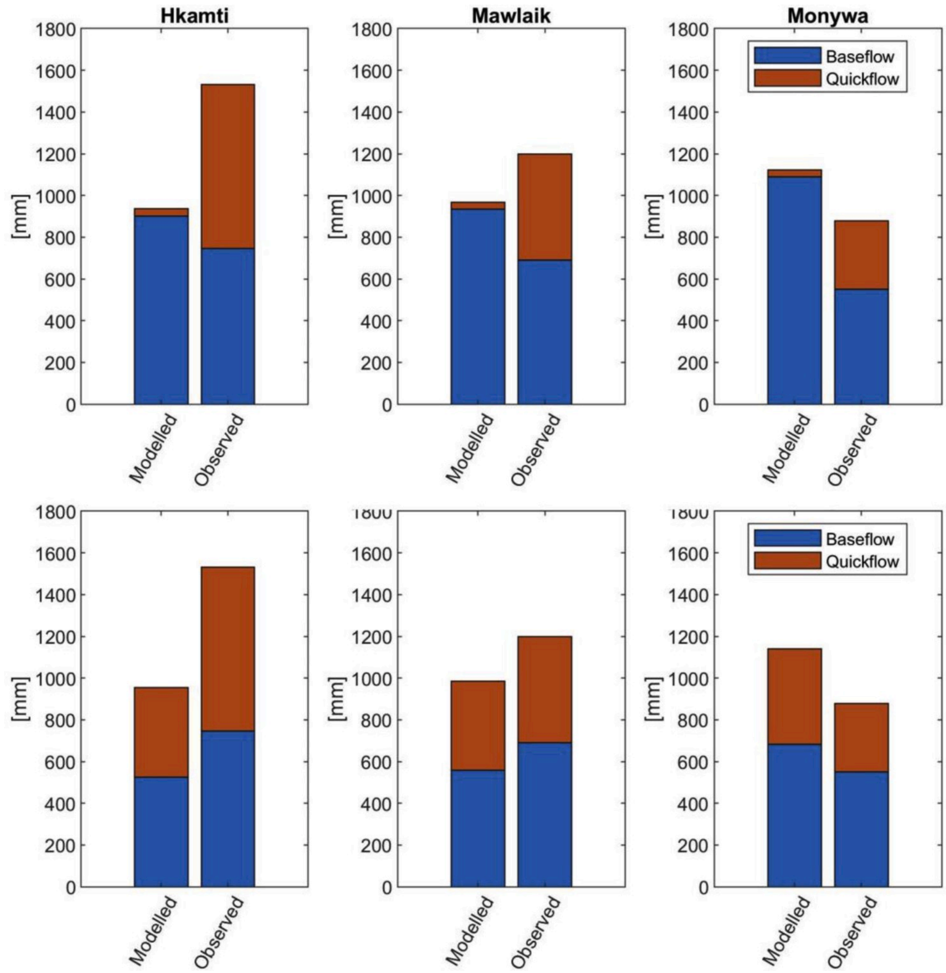

Fig. A1. Comparison of modeled and observed quickflow and baseflow for the three gauging stations for average (top row) and wet (bottom row) antecedent moisture conditions.

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